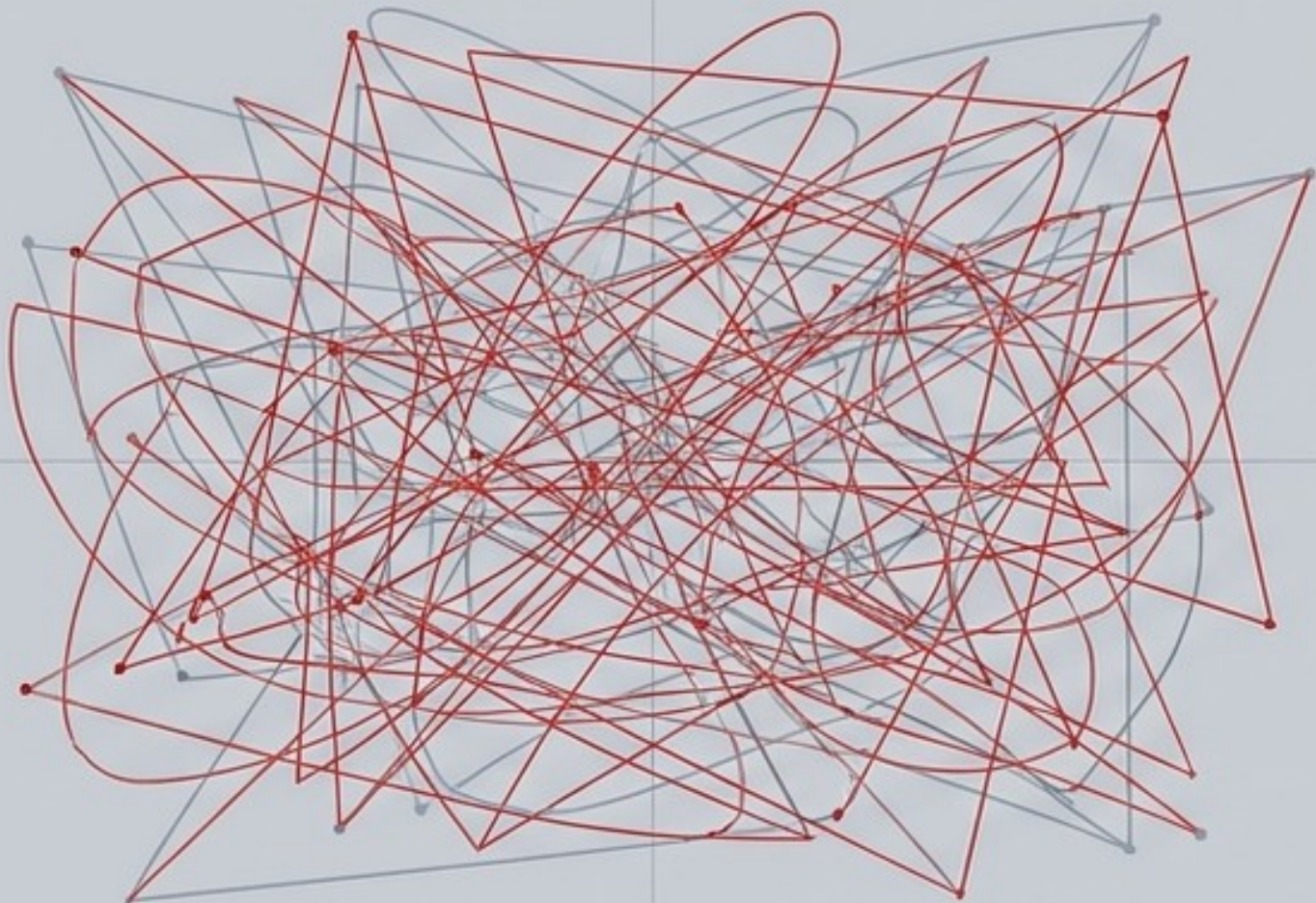


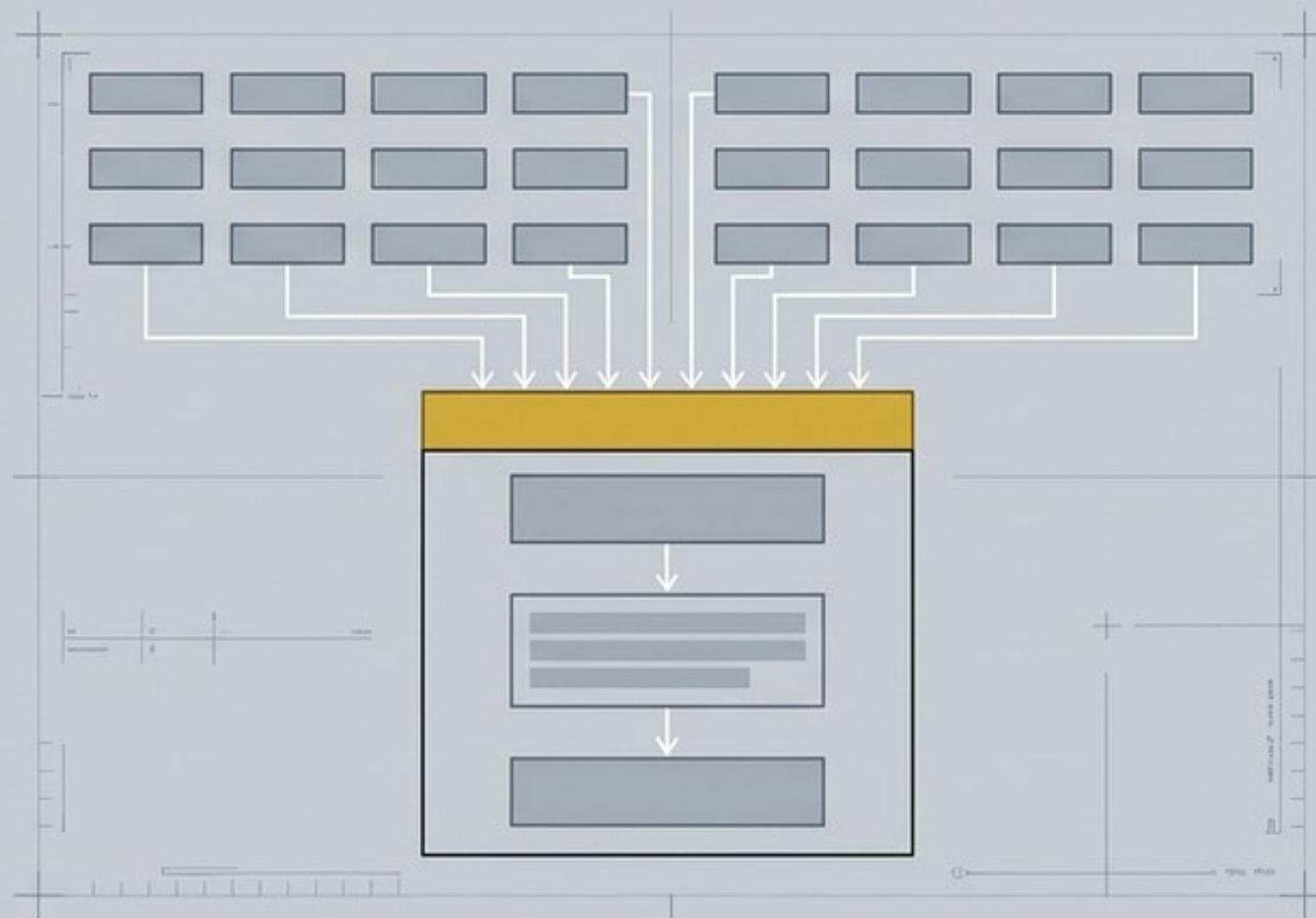
Anatomy of a Block and Block Headers

Architecture, State, and Cryptographic Chaining

Dissecting the internal structure of blockchain data containers, the separation of execution from consensus, and the mathematical metadata that makes history immutable.

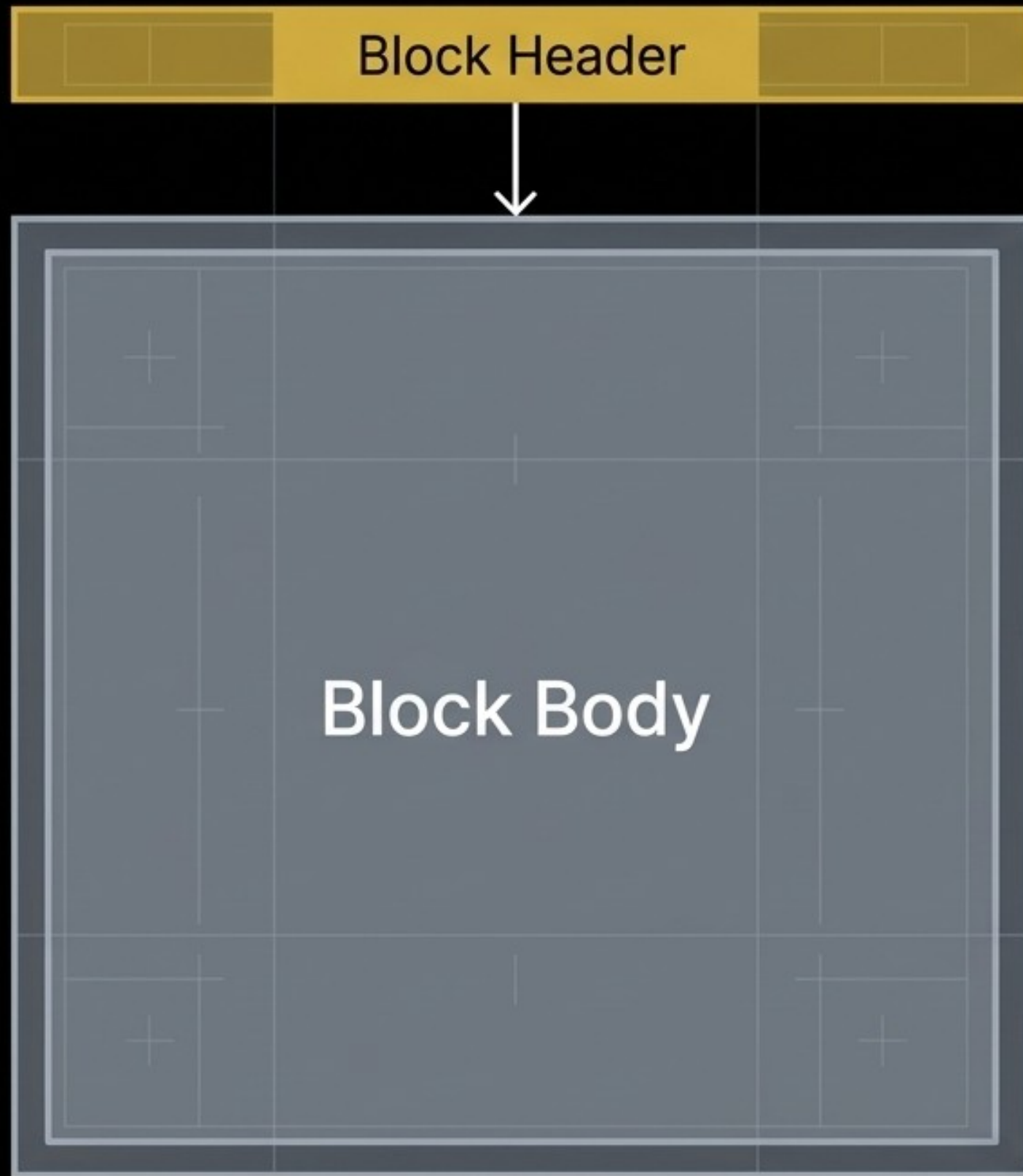


The Individual Gossip Problem: Running global consensus voting for every single atomic transaction triggers endless race conditions and instantly paralyzes network bandwidth.



The Solution: Batching. Grouping thousands of transactions into a single cryptographic batch

- **Consensus Layer:** Agrees strictly on the sequential order of data batches via lightweight metadata.
- **Execution Layer:** Sequentially processes balance mutations from the batch.



Header (Cryptographic Identity)

Highly compact (80 bytes in Bitcoin, ~500–600 bytes in Ethereum).

Acts as the mathematical passport.

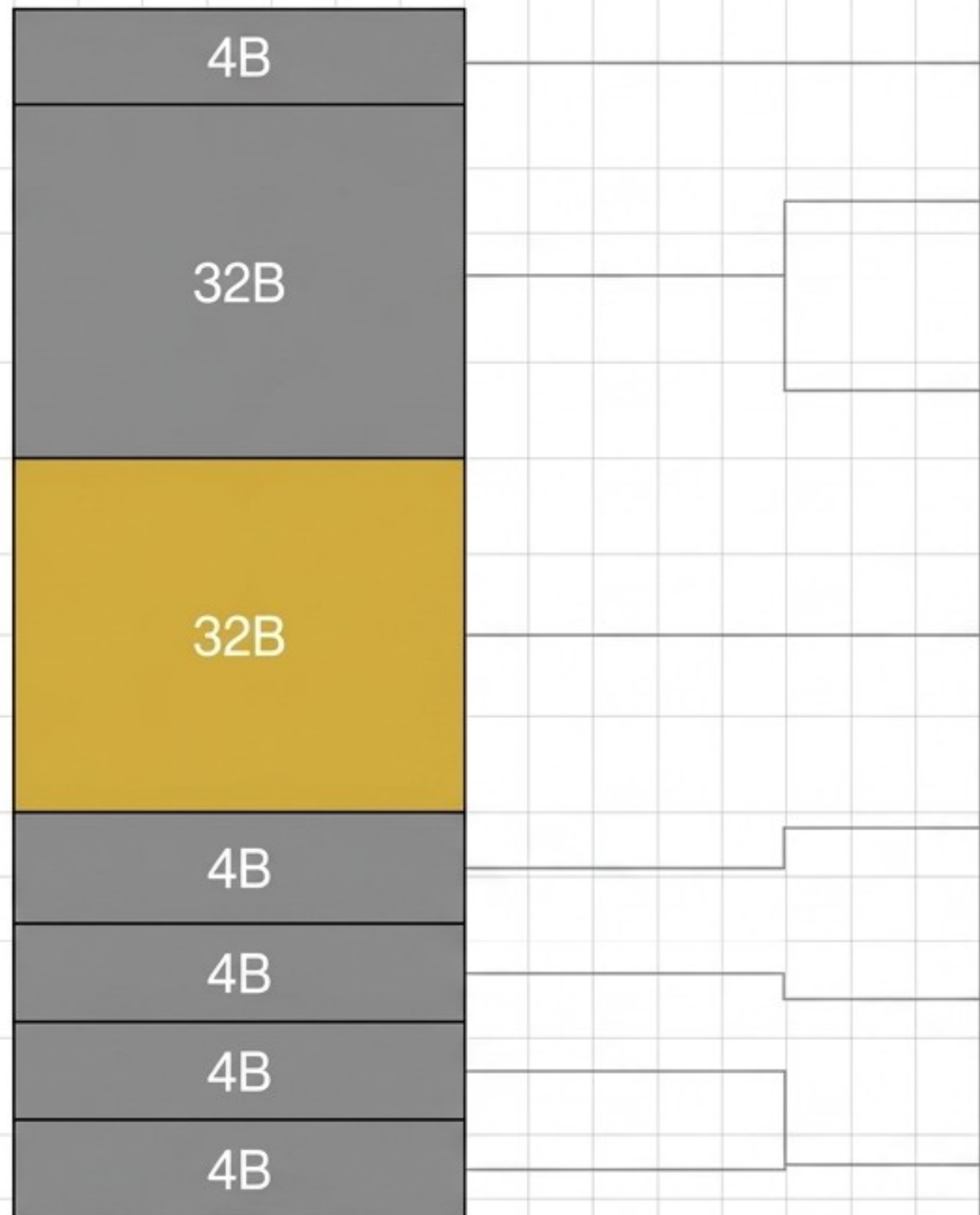
Holds backward hash links, timestamps, and Merkle roots. It is the sole object hashed by miners or signed by validators.

Body (Transaction Payload)

Massive container (1 to 10+ Megabytes).

Holds the raw, user-signed transactions.

Sequentially executed by full nodes to update the database state. Miners do not hash this payload directly.



Block Header Structure (80 Bytes)

nVersion (4 Byte): Protocol version and soft fork upgrade signaling.

hashPrevBlock (32 Byte): Double-SHA-256 hash of the previous header. The backward chain link.

hashMerkleRoot (32 Byte): The cryptographic seal summarizing the entire transaction body payload.

nTime (4 Byte): Unix timestamp of block discovery.

nBits (4 Byte): Compact representation of the Proof of Work difficulty target.

nNonce (4 Byte): 32-bit random counter iterated to find a valid hash.

The nBits Mechanism

Constraint: The difficulty target is a massive 256-bit (32 byte) unsigned integer.

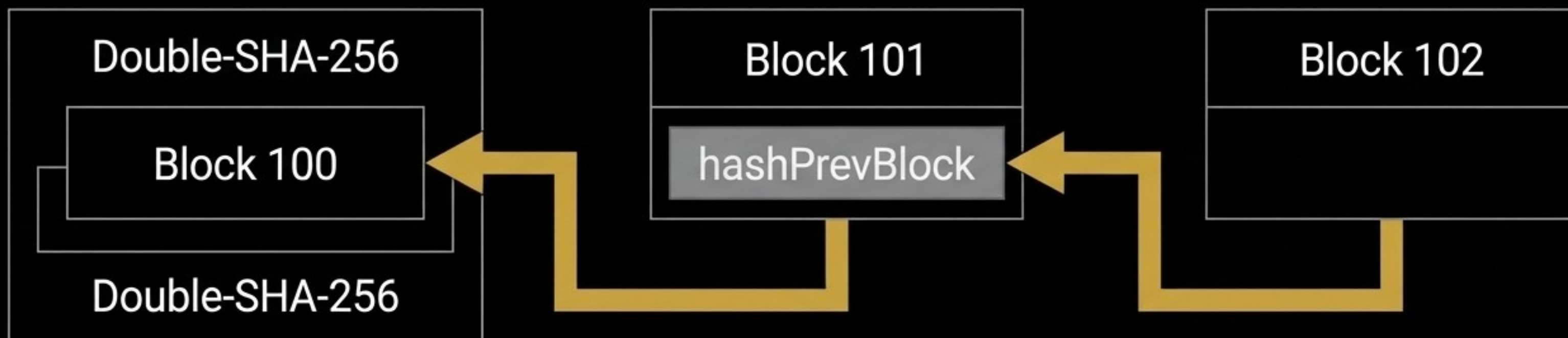
Workaround: nBits compresses this into a 4-byte floating-point format (Exponent + Mantissa). Saves 28 bytes per block worldwide.

$$\text{Target} = \text{Mantissa} \times 256^{(\text{Exponent} - 3)}$$

The nNonce Exhaustion

Constraint: The 4-byte nNonce only provides 2^{32} (4.29 billion) combinations. Modern ASICs exhaust this entire 32-bit search space in just 40 microseconds.

Workaround: Mutating the ExtraNonce (a 4-8 byte counter inserted into the Coinbase transaction scriptSig). This alters the Coinbase hash → completely resets the hashMerkleRoot in the header → refreshes the 32-bit nonce space.

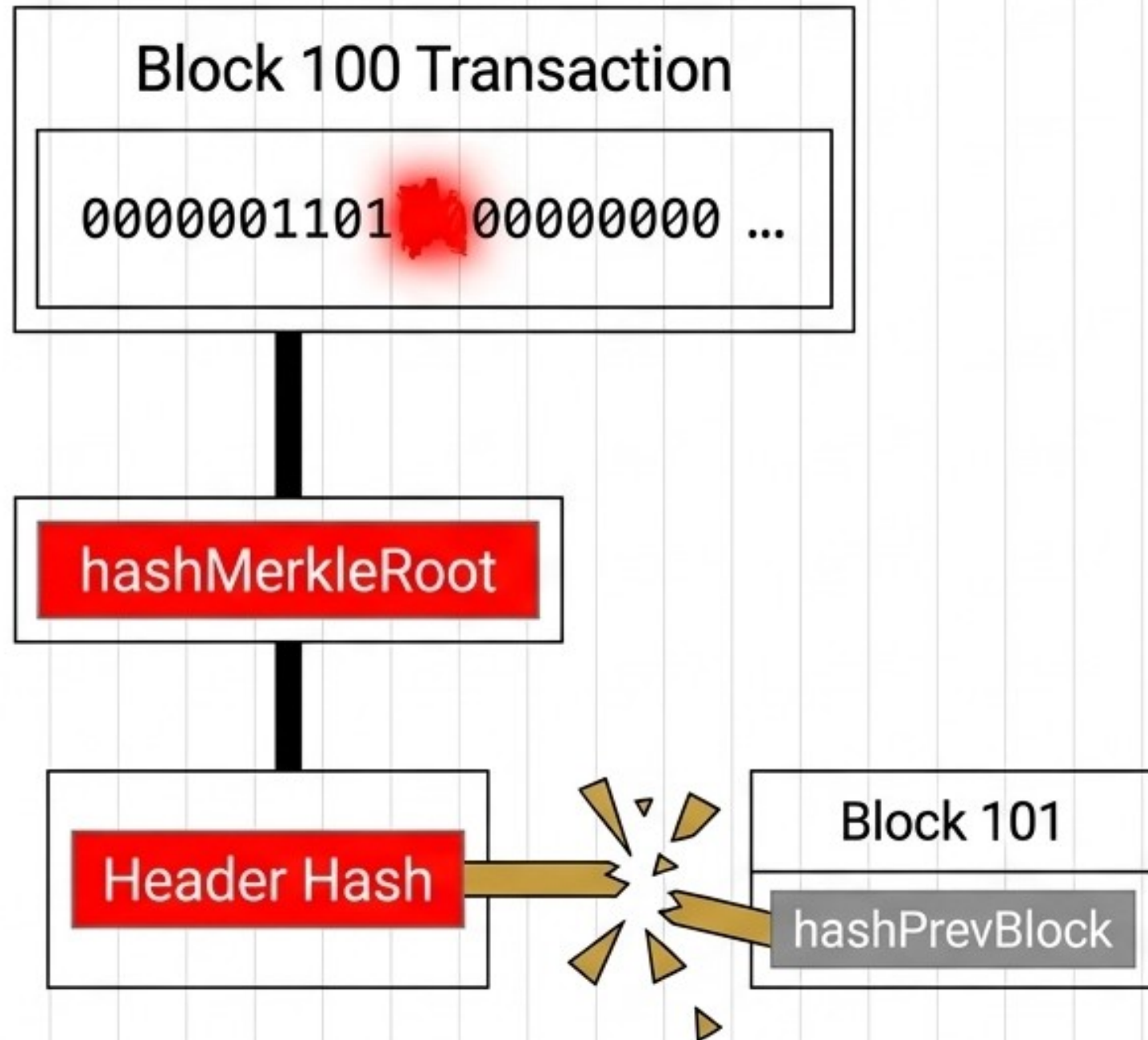


$$\text{Hash}(B_n) = \text{SHA-256}(\text{SHA-256}(\text{Header}_n))$$

The Deterministic Dependency Tree.

Every block explicitly embeds the cryptographic signature of its predecessor. Because the hash relies on the entire header, each block mathematically seals the complete history of the network all the way back to the Genesis Block.

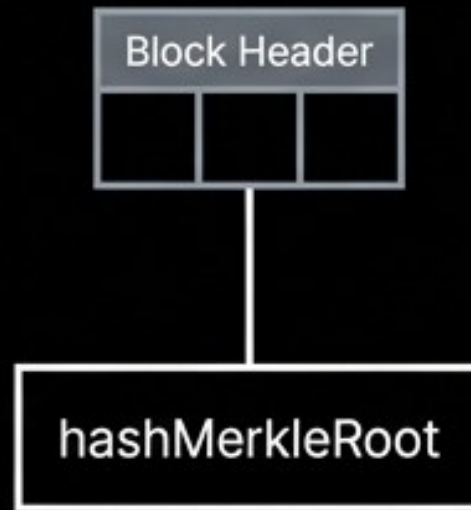
The Mallory Attack



1. Mallory alters 1 byte of historical transaction data to steal coins.
2. The transaction hash instantly changes, mutating the hashMerkleRoot.
3. The altered Merkle Root alters the total Header Hash of Block 100.
4. **Fatal Impact:** The cryptographic link is severed. Block 101's hashPrevBlock pointer fails. Honest nodes instantly reject the chain. **The attacker is forced to out-compute the entire global network to rebuild the chain from the point of fracture.**

Bitcoin

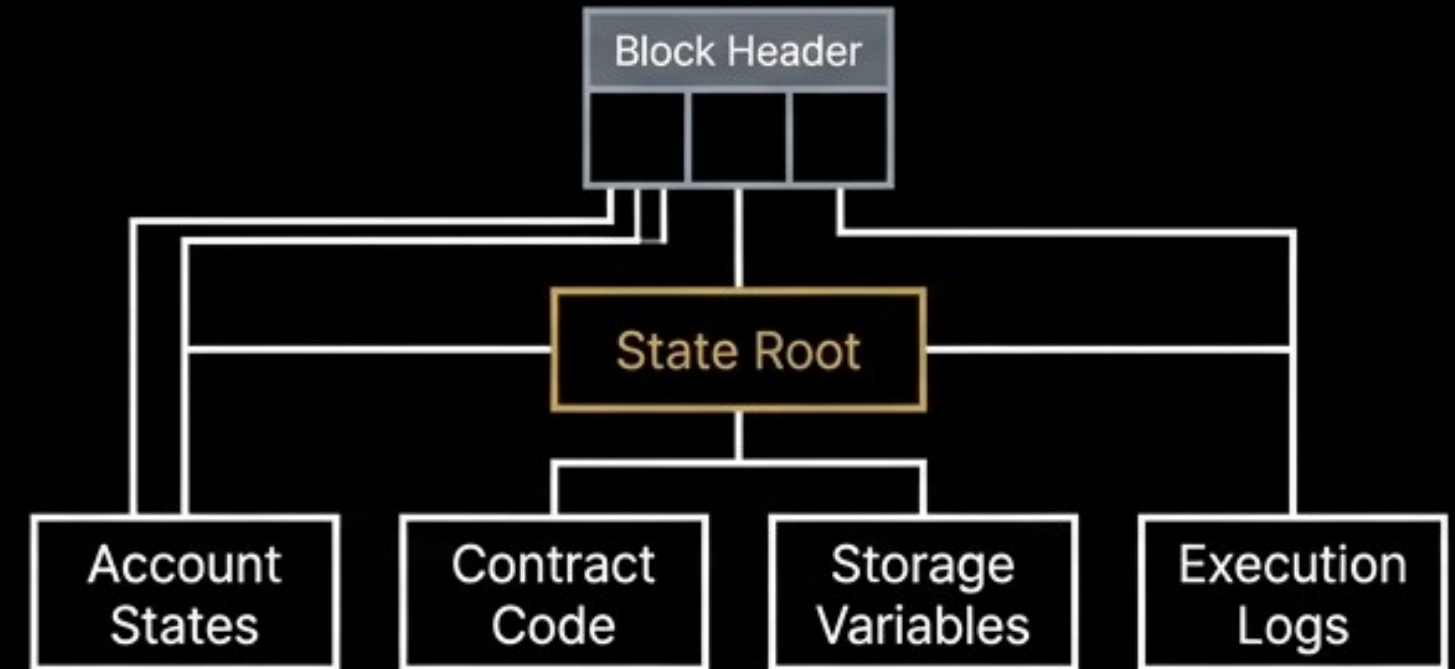
Static Ledger



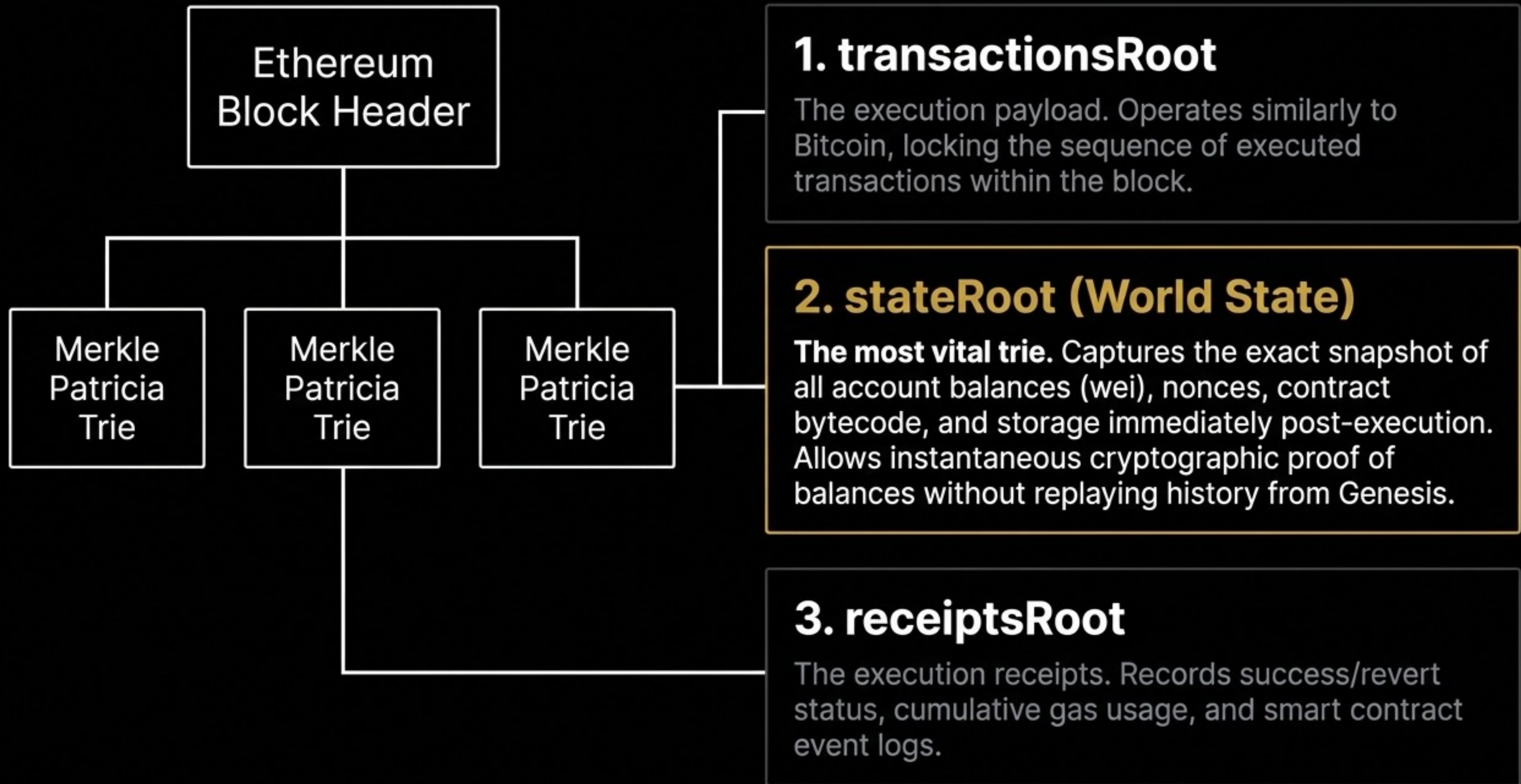
- Tracks simple coin mutations.
- Minimal metadata.
- Header size: Exactly 80 bytes.
- Single Merkle root locking the transaction payload.

Ethereum

Global Compute



- Turing-complete state machine.
- Tracks account balances, contract bytecode, storage variables, and execution logs.
- Header size: ~500 to 600+ bytes.
- Requires advanced computational metadata and dynamic fee algorithms.



Advanced Execution Metadata

logsBloom (256 Byte) Probabilistic Bloom filter for instant event log querying without downloading massive receipt tries.	gasLimit & gasUsed Computational resource meters restricting workload size per block to protect validator CPUs.
baseFeePerGas (EIP-1559) Algorithmic dynamic base fee that is automatically burned upon transaction.	prevRandao Verifiable pseudo-random entropy injected directly from the Beacon Chain (Proof of Stake consensus).

The Next Step



The Container is Built. How does the data arrive?

We have mapped the cryptographic vault. But how does a user's intent traverse node RPCs, navigate the mempool, survive the Dark Forest of front-running, and execute deterministically?

Next Module: Transaction Lifecycle and State Transitions.